

Doc. 308 – The Cosmic Sector of FFGFT:
Redshift, Dark Sector and
Geometric Gravity

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Doc. 308 – The Cosmic Sector of FFGFT

What this is about

FFGFT advances an unusual claim: the universe is not expanding. Cosmic redshift is not a Doppler effect but a geometric property of the ξ -field. Dark energy does not exist as a physical entity. And dark matter requires no particles — its effects are to follow from the same geometry that carries light deflection; this is established for gas-dominated dwarf galaxies, for spiral galaxies and the Bullet Cluster it is open.

This document assembles the cosmic sector of FFGFT: what has been derived, what has been observationally confirmed, and where the limits lie. It addresses readers with a physics background who want to know *what* the theory says and *why*.

The three sources of degeneracy are named, the two cores of the geometry are developed, and the status of each step — derivation, structural argument, posit — is declared.

Conventions

Legend.

- [K]** *Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
- [S]** *Structural claim* — justified but not yet fully proved.
- [POSIT]** *Posit* — declared assumption, not further derived.

Central quantities.

$\xi = 4/30000$	Only fundamental parameter of FFGFT [POSIT, P33].
$\tilde{T} \cdot m = 1$	Foundational relation: intrinsic time and mass are inversely coupled.
$R_H = c/H_0$	Geometric maximal scale — not an expansion horizon.
$a_0 = c^2 \xi^{10} / (4\bar{\lambda}_e)$	Transition scale of the inertia regime; below a_0 inertia is non-linear ($\tilde{T} \cdot m = 1 + \text{Unruh}$); $= 1.033 \times 10^{-10} \text{ m/s}^2$.
$\bar{\lambda}_e = \hbar/(m_e c)$	Reduced Compton wavelength of the electron.

Chapter A

The Static Universe and Redshift

A.1 The fundamental choice

Λ CDM describes an expanding universe: galaxies move apart, light is redshifted by the expansion of space, and the observation of distant supernovae reveals an accelerated expansion accounted for by a cosmological constant Λ .

FFGFT makes a different fundamental choice: the universe is static. Cosmic redshift arises not from motion but from geometric path elongation in the ξ -field along the light path.

A.2 The redshift mechanism [K]

The FFGFT redshift formula (Doc. 280):

$$1 + z = e^{\xi x} \quad \implies \quad z \approx \xi x \quad (\text{for small } z),$$

where x is the path length, measured in a length that is not derived from ξ and is fixed only via the measured H_0 : the form of the relation is derived, its scale is calibrated (P39). Each photon loses energy along its path through fractal path elongation — proportional to distance, without wavelength dependence.

Achromatic [K]: The path elongation is purely geometric and stretches *all* wavelengths by the same factor $e^{\xi x}$. Finite-element simulations of the geometry confirm this (K6, Doc. 190). A chromatic redshift would destroy the coherence of the CMB spectral shape — this is not observed.

A.3 The degeneracy of the data [K]

For the three classical pillars of cosmology — Type Ia supernovae, baryon acoustic oscillations (BAO), CMB temperature — the following holds: they cannot in principle distinguish between expansion and static geometric path stretching, as long as both models produce the same distance-redshift relation $z(d)$ (Doc. 267).

This is not a weakness of FFGFT. It is a methodological statement: *these data alone do not decide between the models*. The decision shifts to theoretical consistency, parameter count and derivability.

A.4 What follows from this

For dark energy: Λ is needed in Λ CDM to account for the apparent accelerated expansion. In the static reading there is no expansion to be accelerated. Λ loses its referent in the static reading — it is not falsified, it becomes objectless.

For the Hubble tension: Different measurement methods for H_0 give different values. In the FFGFT reading, H_0 does not measure an expansion rate but a parameter of the ξ -geometry. FFGFT knows only *one*, calibrated H_0 value and does not distinguish between early and late measurements; the tension cannot be modelled in FFGFT — it is neither reproduced nor resolved.

For the z -axis: Every presentation of cosmological data “over time” is formulated in Λ CDM language. An FFGFT interpretation must treat z as a geometric distance parameter, not a time axis.

Chapter B

Geometric Gravity: the Two-Halves Synthesis

B.1 Einstein 1911 and 1915

Einstein calculated the deflection of light at the Sun in 1911 from time dilation alone:

$$\alpha_{1911} = \frac{2GM}{c^2 b} = 0.875''.$$

That was half the measured value. The 1915 theory of GR yielded the full value $\alpha_{1915} = 1.750''$, because space *and* time curvature each contribute one half.

FFGFT *explains* this factor of 2 from first structure.

B.2 K3 area invariance [K]

$\tilde{T} \cdot m = 1$ (time curvature) and fractal path elongation (space curvature, Doc. 182) are in FFGFT not alternative mechanisms but **complementary halves**. The K3 rule — area invariance of the time-space 2-cell — anchors this:

$$L_t \cdot l = \text{const.}$$

This is formally the reciprocity form of $\tilde{T} \cdot m = 1$, applied to the time-space pair.

FE verification [K]: Ray tracing through three configurations (30001 log nodes, RK4, `ffgft_308_p36_stufeB1_faktor2_fe.py`):

Configuration	α at solar limb	Ratio to 1.75''
Time rate only ($\tilde{T} \cdot m = 1$)	0.875''	0.500
Path elongation only (space)	0.875''	0.500
Both together (K3)	1.750''	1.000

PPN test [K]: K3 is the sole survivor of the T^4 selection computation (four candidates tested, `ffgft_308_p36_stufeB2_kappa_herleitung.py`): $\gamma_{\text{PPN}} = 1$ exactly. Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ passed with unlimited margin.

B.3 The same structure in the inertia regime: a_0 [K]/[S]

The K3 two-halves structure appears a second time, at cosmological scale. From Doc. 279 (the exponent 10 is calibrated to H_0 , P20/P39; Doc. 309):

$$\frac{H_0}{c} = \frac{\pi}{2} \frac{\xi^{10}}{\bar{\lambda}_e} \quad \Rightarrow \quad R_H = \frac{2 \bar{\lambda}_e}{\pi \xi^{10}}.$$

R_H is the geometric maximal scale of the compact T^4 geometry (Doc. 182) — not an expansion horizon, valid in the static reading.

The cycle doubling [S]: The K3 2-cell has two cycles. A closed path that traverses each once measures:

$$L = \underbrace{2\pi R_H}_{\text{time cycle}} + \underbrace{2\pi R_H}_{\text{space cycle}} = 4\pi R_H.$$

Rindler saturation [S]: The Rindler horizon of an observer with acceleration a lies at c^2/a . When it reaches L , the acceleration is no longer relationally definable. This sets:

$$a_{\text{bg}} = \frac{c^2}{4\pi R_H} = \frac{c^2 \xi^{10}}{8 \bar{\lambda}_e}.$$

Inertia from $\tilde{T} \cdot m = 1$ [S]: The foundational relation is a statement about mass. If inertia follows the time field (Unruh coupling), the force law in closed form is:

$$\boxed{a = \sqrt{a_N^2 + a_N a_0}} \quad \text{with} \quad a_0 = 2 a_{\text{bg}} = \frac{c^2 \xi^{10}}{4 \bar{\lambda}_e} = 1.033 \times 10^{-10} \text{ m/s}^2.$$

The 2π of the geometry cancels exactly against the $\pi/2$ from Doc. 279. The rational factor 4 remains:

$$4 = \underbrace{2}_{\text{K3: time and space cycle}} \times \underbrace{2}_{\text{Unruh: } a_0=2a_{\text{bg}}}.$$

No step contains a further free parameter; the scale takes over the calibration of the exponent 10 to H_0 .

Chapter C

The Dark Sector

C.1 Dark energy: not a physical object

Λ arises in Λ CDM as an answer to the apparent accelerated expansion. In the static reading this expansion does not exist. Λ thereby loses its referent — it is not falsified by the reading change, it becomes objectless.

Three conditions speak against it:

1. **Only from degenerate data.** SNe Ia, BAO, CMB — exactly the datasets between which both readings are indistinguishable.
2. **No independent anchor.** The only theoretical prediction (QFT vacuum energy) misses the value by 10^{122} .
3. **Objectless under reading change.** Not refuted, but without referent.

That is the status: reading artefact. Not “dark energy does not exist” — that would be a statement within the expansion reading — but: it is a reading-internal bookkeeping quantity.

C.2 Dark matter: a regime, not a substance

The findings — rotation curves, gravitational lensing, Bullet Cluster — are reading-independent. They do not disappear under a reading change. FFGFT traces them, without a new ingredient, to the same force law that $\hat{T} \cdot m = 1$ together with K3 geometry delivers; this is established for the gas-dominated dwarf galaxy DDO 154, for spiral galaxies and the Bullet Cluster it is open (see below).

Dark matter in FFGFT is not a substance. It is a name for a regime.

The force law in the galactic regime [K]/[S]:

$$a = \sqrt{a_N^2 + a_0 a_N}.$$

For $a_N \gg a_0$ (inner regions): $a \rightarrow a_N$ (Newton). For $a_N \ll a_0$ (outer regions): $a \rightarrow \sqrt{a_N a_0}$, hence $v^4 = GM_b a_0$ — the baryonic Tully-Fisher relation.

In FFGFT, dark matter does not denote a missing mass component but the regime below a_0 in which inertia is non-linear. The missing mass is not missing — inertia is different there.

Results [K] (ffgft_308_p36_stufeA_a0_anker.py):

Galaxy	Type	v_{Newton}	v_{FFGFT}	v_{obs}
DDO 154	dwarf, gas-dominated	18.5	47.1	47.0 km/s
NGC 3198	spiral	50.4	121	150 km/s
Milky Way	spiral	122	180	220 km/s

DDO 154 is the cleanest test (gas-dominated, M_b best known): $v_{\text{FFGFT}}/v_{\text{obs}} = 1.002$. The deviations for NGC 3198 and the Milky Way ($121/150 - 1 = -19.3\%$, $180/220 - 1 = -18.2\%$) lie well outside the M_b uncertainty: for $v \propto M_b^{1/4}$, $\pm 30\%$ in M_b correspond to only $+6.8\% / -8.5\%$ in v (even in the Newtonian limit only $+14\%$). Closing them would require $M_b \times 2.36$ (NGC 3198) and $\times 2.23$ (Milky Way). For spirals the test is open.

Bullet Cluster [K] (ffgft_308_p36_stufe01_bullet_cluster.py): Two-clump FE projection (Clowe+2006 parameters) at instantaneous coupling. According to Clowe et al. 2006 the X-ray gas is the dominant baryonic component ($M_{\text{gas}}/M_{\text{stars}} \approx 5-10$; gas $\sim 85\%$ of the baryons, i.e. $5.7 : 1$). With these baryonic masses the lensing peak lies at 240 kpc, i.e. at the gas; it moves to the gas already from gas:stars ≈ 1.4 . A galaxy mass of $2.3 \cdot 10^{14} M_{\odot}$ would be the lensing mass including the missing component; a test with it is circular. The Bullet Cluster is therefore not established as passed; the test is open.

C.3 Asymmetry in comparison

ξ is cross-anchored — it carries the lepton mass ladder, the Koide scalar, the α reference, K_{frak} and k^* (Doc. 275). It works in five foreign sectors before the cosmological one.

Δ works nowhere except its own fit.

That is the difference: not "true vs. false" but "cross-anchored vs. isolated".

Chapter D

Summary: Picture of the Cosmic Sector

Quantity	FFGFT	Status
Cosmic redshift	$1 + z = e^{\xi x}$, scale via H_0	[K] (form)
Achromatic	geometric, FE confirmed	[K]
Degeneracy SNe/BAO/CMB	between both readings	[K]
Λ (DE)	reading artefact, no referent	[K]
Hubble H_0	$c\pi\xi^{10}/(2\bar{\lambda}_e)$, exponent calibrated	[POSIT]
Light deflection	$0.875'' + 0.875'' = 1.750''$	[K]
$\gamma_{\text{PPN}} = 1$	K3 sole survivor	[K]
$a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$	chain via calibrated H_0	[K]
Rotation curves	DDO 154 ratio 1.002; spirals ~18 % low	[K]; spirals open
Bullet Cluster	baryonic: peak at the gas (240 kpc)	open
K3 cycles $4\pi R_H$	time + space cycle equal	[S]
Rindler saturation	horizon meets maximal scale	[S]
$\xi = 4/30000$	magnitude	[POSIT]
K3 forward proof	2-cell vs. 4-volume	open
14% to a_0^{emp}	literature range $1.1\text{--}1.3 \times 10^{-10}$	open

The cosmic sector of FFGFT rests on a single mechanism: $\tilde{T} \cdot m = 1$ together with K3 area invariance produces light deflection, geometric gravity and achromatic redshift. a_0 follows from the same chain without a further free parameter; the scale carries the calibration of H_0 .

Dark energy is not a physical object — it is a construct of the expansion reading. Dark matter as particles is not required insofar as the effects follow from the geometry — established for gas-dominated dwarf galaxies, open for spiral galaxies and the Bullet Cluster.

Chapter E

References

Internal documents: Doc. 025 (cosmology overview); Doc. 026 (FE geometry); Doc. 064 (redshift foundations); Doc. 182 (maximal scale); Doc. 267 (cosmological degeneracy); Doc. 279 (H_0 from ξ^{10} chain, K6); Doc. 280 (achromatic redshift); Doc. 275 (k^* recursion); Doc. 190 (P33, P44, K6).

Verification scripts (all inputs declared): `ffgft_308_p36_stufeA_a0_anker.py` (chain $\xi \rightarrow a_0$, rotation curves); `ffgft_308_p36_stufeB1_faktor2_fe.py` (factor-2 gate, 30001 log nodes, RK4); `ffgft_308_p36_stufeB2_kappa_herleitung.py` (selection computation K3); `ffgft_308_p36_stufeC1_bullet_cluster.py` (Bullet Cluster projection).

External references: McGaugh, Lelli, Schombert 2016, PRL 117, 201101 (RAR/ a_0); Clowe et al. 2006, ApJ 648, L109 (Bullet Cluster).